

Landscape Simulation Exercise

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Simulating Landscape Changes

The present game-exercise intends to give you a hands-on experience on how ecological modelers use data, statistics, and probability to forecast changes in our planet. To play the game, you were previously divided in three groups:

1. the **Turtle** group
2. the **Bird** group, and
3. the **Tree** group.

If you are reading this, you belong to the Tree group, which we will rename to ***Landscape Simulation*** group. Fancier, right? :)

You will be responsible for simulating changes in the landscape and, ultimately, influencing how the turtles and birds populations will change.

In the folder you downloaded, you received a spreadsheet named `landscape_simulation.xlsx`. This sheet brings the aerial view of your landscape and you can see how it changes through time.

1. Getting started

Before you start, you will need to install three libraries to help with the simulations, if you haven't done it already. The first library is called **Require**, and is an R package (or library) dedicated to improve code reproducibility on the setup side. The second library we will need is called `data.table`, which improves efficiency of data manipulation. The third is a library to deal with spatial analyses.

```
if (!require("Require")) {install.packages("Require"); require("Require")}  
Require("data.table")  
Require("terra")
```

The next thing we will do is to **source** (or in other words, *read and be aware of it*) all **functions** that will help us simulate the changes in the landscape. These functions are described in the file `landscapeSimulation.R`, which is a typical coding file for the R environment. Functions are the basis for programming. They are generally generic commands to explain to the software what to do. If you are interested, you can read more about:

R and functions, in this great and simple tutorial from Norm Matloff: <https://github.com/matloff/fasterR> and in a cool YouTube video from R Programming <https://www.youtube.com/watch?v=BvKETZ6kr9Q>

data.table and efficient data manipulation in this tutorial: <https://www.machinelearningplus.com/data-manipulation/datatable-in-r-complete-guide/>

RMarkdown and other awesome packages from Yihui Xie here: <https://bookdown.org/yihui/rmarkdown/> and <https://bookdown.org/yihui/rmarkdown-cookbook/>

RStudio in a fun and easy way: <https://moderndive.netlify.app/1-getting-started.html>

The present `landscapeSimulation.Rmd`, on the other hand, is a mixed-file that allows for both documentation (normal text) and code to be written. When the file is done, this document can be **rendered** into a PDF

with a really neat format, which you can also see in the folder, as `landscapeSimulation.pdf`.

2. Functions' Sourcing

To **source** the functions we will using, we can use a function in R called `source()` (I agree, not too creative, but easy to remember). To run the line below, you can press the little green arrow on the right side of the **code chunk**, where the instructions for R to source the file with functions are written.

```
source("landscapeSimulation.R")
```

If everything goes according to the plan, you should have seen in the **Console** below this window the following sentence:

All functions were correctly sourced! You are ready to start.

Now the real fun begins!

3. Possible fates for different landscape types

For each 10 year time step, we will simulate the fate of 1. Forests 2. Human disturbance

To make it fun and understandable, we will do a very (VERY!) simple exercise on simulation, where water bodies are kept constant through time, climate change is ignored (although it should NOT in real life), and the processes for all animals and landscape a very simplified. After the exercise we will discuss a study that brings in considerably more complexity. But one thing at a time.

First, we will discuss how each element can be changed. Then, we will start changing them.

3.1. Forests

There are two important things about forests in our game that you need to remember:

1. Forests have one very important property for animals: **age**.
2. Forests can have only three behaviors: **grow**, **burn** or be **replaced by human development**.

In our game, older forests are less resilient to fire, followed by middle-aged forests, and then by young forests. If forests do not burn nor are replaced by human development, they will grow. A forest that is 10 years old today will be 20 years old in 10 years, the time step we are using for our landscape simulations.

3.2. Human disturbance

There is only one level of human disturbance, which can be grey or black. The size of new developments will always be 1. Human disturbance can only **appear** in a *forested area* or in a *burned area*, being more likely in a burned area, followed by a *young forest*, *middle-aged forest* and only then by *older forest* (in our game, everyone understand and respect the importance of forests!).

4. Simulating landscape changes

First of all, we will create a landscape where the changes will happen:

```
landscape <- getInitialLandscape()
```

Here we can see the legend of our landscape: 1. Old forest (dark green) 2. Middle-aged forest (middle green) 3. Young forest (light green) 4. Recently burned patch (light yellow) 5. Human disturbed patch (grey or black) 6. Water (light blue)

Once we have our landscape, we will have changes happening in the following order: 1. Forest growth 2. Human development 3. Fire


```
= 3)  
simulateFire(intensity = 0.5)
```

I hope you had some fun, but most of all, could grasp the general mechanisms behind how we simulate landscape changes.